
MA 551: Numerical Analysis

Course Code	MA 551
Course Name	Numerical Analysis
L-T-P-C	3-0-0-3
Prerequisites	IC 110 Engineering Mathematics, IC 111 Linear Algebra
Intended for	3rd and 4th year UG/PG
Distribution	Elective
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **Introduction** [6 Lectures]

Approximate Numbers and Significant Digits, Propagation of errors, Different types of errors, Backward error analysis, Sensitivity and conditioning, Stability and accuracy.

- **Nonlinear Equations** [8 Lectures]

Bisection method, Newton's method and its variants, Secant method, Fixed point iterations and their Error analysis.

- **Module III** [8 Lectures]

Finite differences, Polynomial interpolation, Newton Divided Differences, Spline interpolation. Numerical integration, Trapezoidal and Simpson's rules, Newton Cotes formula, Gaussian quadrature, and Numerical differentiations.

- **System of linear equations** [8 Lectures]

Gaussian Elimination, Partial Pivoting, Pivoting and Scaling in Gaussian Elimination method, Iteration methods, Error analysis.

- **Initial Value Problem (IVP)** [6 Lectures]

Taylor series method, Euler and modified Euler methods, Runge Kutta methods, Multistep methods, Predictor Corrector method.

- **Boundary Value Problem (BVP)** [6 Lectures]

Text Books

1. K. E. Atkinson, **An Introduction to Numerical Analysis**, 2nd Edition, John Wiley, 2008.
2. S. D. Conte and Carl de Boor, **Elementary Numerical Analysis**, McGraw Hill, 1988.

References

1. M. T. Heath, **Scientific Computing: An Introductory Survey**, McGraw Hill, 2002.
2. Ralston and P. Rabinowitz, **A First Course in Numerical Analysis**, Dover Publications, 2001.